Developmental Foundations of Tangible Interactive Learning Systems for Early Childhood:

The Böðn Project

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Abstract

Böðn is an open-source, battery-powered tangible learning system designed for children aged 3–7. The device combines physical controls (buttons, rotary encoders, toggle switches) with multimodal feedback (LED animations, synthesized audio, screen graphics) to deliver play experiences that target core executive functions—working memory, inhibitory control, and cognitive flexibility—as well as sensorimotor, cognitive, language, and social-emotional development. This paper reviews the developmental science that informs Böðn's design, maps each feature to established developmental domains, and identifies areas for future development. We argue that tangible interfaces offer advantages over touchscreen-only systems for early childhood learning, and that game-based, guided play can meaningfully support executive function development during the critical preschool window.

Keywords: executive functions, tangible user interfaces, early childhood, guided play, physical computing, open-source hardware

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1 Introduction

The preschool years (ages 3–7) represent a critical period for the development of executive functions (EF)—the cognitive control processes that enable goal-directed behaviour (Diamond, 2013). A large body of evidence links early EF to school readiness (Blair and Razza, 2007), social competence, and lifelong well-being (Moffitt et al., 2011). Unlike IQ, which is relatively stable, executive functions are highly malleable during early childhood and respond to targeted practice (Diamond, 2013; Montroy et al., 2016).

At the same time, the proliferation of touchscreen devices has raised concerns about passive screen time in young children (Radesky et al., 2015). The World Health Organization recommends no more than one hour of sedentary screen time per day for children aged 2–4 (World Health Organization, 2019). This creates a tension: digital systems can deliver adaptive, engaging practice that physical toys cannot, yet the medium itself may undermine the developmental benefits.

Tangible user interfaces (TUI)—physical objects that bridge the digital and physical worlds—offer a potential resolution. Since Ishii and Ullmer (1997) introduced the concept, a growing literature has explored TUI for educational contexts, finding advantages in engagement, spatial understanding, and embodied learning (Marshall, 2007; Price et al., 2003).

This paper presents **Böðn**, an open-source tangible learning system that combines physical controls with adaptive, game-based experiences designed around the developmental science of early childhood. We review the theoretical foundations, describe the system design, map each feature to developmental domains, and identify directions for future work.

2 Background

2.1 Executive Functions in Early Childhood

Miyake et al. (2000) proposed the influential three-factor model of executive functions: *working memory* (updating), *inhibitory control* (inhibition), and *cognitive flexibility* (shifting). While originally validated in adults, subsequent work has confirmed this structure in preschoolers, with important developmental nuances (Montroy et al., 2016).

Working memory is the ability to hold and manipulate information in mind. In 4-year-olds, typical capacity is 2–4 items (Diamond, 2013). Tasks like remembering a sequence of coloured lights and reproducing it from memory directly exercise this capacity.

Inhibitory control encompasses both response inhibition (suppressing a prepotent motor response) and interference control (resisting distracting information). The classic "day–night" task, where children must say "night" when shown a sun card, illustrates the

challenge for preschoolers (Diamond, 2013).

Cognitive flexibility requires disengaging from one rule set and engaging a new one. The Dimensional Change Card Sort (DCCS) task (Zelazo, 2006) is the standard assessment: children sort cards first by colour, then by shape. Most 3-year-olds fail the switch; most 5-year-olds succeed. The transition period (ages 3–5) represents a critical window for practice.

Longitudinal data confirm that self-regulation (a broader construct overlapping substantially with EF) improves steeply between ages 3 and 5, with individual differences in developmental trajectory already predicting kindergarten readiness (Montroy et al., 2016). The practical implication is that *when* intervention happens matters: the same dose of practice delivered at age 3–5 is likely to produce larger gains than at age 7+.

The EF-training literature itself is mixed. Diamond and Ling (2016) reviewed dozens of intervention studies and reached a sobering conclusion: computerised working-memory training generalises poorly beyond the trained task, while play-based, embodied, and socially embedded approaches (martial arts, mindfulness, curricula such as Tools of the Mind) produce broader and more durable transfer. The authors identify several common features of programs that work — sustained engagement, challenge calibrated to the child's current level, multimodal practice, and low-stakes failure. Böðn's design is explicitly shaped by this evidence: every game mode is playful rather than drill-like, difficulty scales adaptively, feedback is multimodal, and errors are gentle.

2.2 Guided Play and Playful Learning

Weisberg et al. (2013) define guided play as adult-scaffolded activities where children explore within a structured environment designed to make discovery productive. Guided play occupies a middle ground between free play (child-directed, no adult structure) and direct instruction (adult-directed, no child agency). Evidence suggests that guided play produces deeper learning than either extreme for concept acquisition and problem-solving (Hirsh-Pasek et al., 2015).

Skene et al. (2022) reported the most comprehensive quantitative evidence to date: a systematic review and meta-analysis of 39 studies covering 3,800 children, finding that guidance during play produced greater gains than both free play and direct instruction for early mathematics, shape knowledge, and task-switching (an executive-function measure). The effect was strongest when the child retained agency over how the activity unfolded even while an adult (or an environment) set the frame. Hassinger-Das et al. (2017) extend this framework to well-designed games, arguing that games sit naturally on the guided-play continuum: they supply clear goals and rules while preserving voluntary engagement and child-led choice within those rules. Böðn's game modes are deliberately structured this way — each mode has a bounded rule system (the "frame"), but within the frame the

child chooses when to play, how fast to progress, and what to explore.

This framing also helps explain why open-ended modes like Mystery Box and Tone Lab sit alongside rule-based modes like Simon and Rule Follow without contradiction. Both are species of guided play: the environment's structure differs, but in each case the child's action remains voluntary and self-directed.

2.3 Tangible User Interfaces for Young Children

Ishii and Ullmer (1997) introduced the concept of tangible user interfaces as a way to "give physical form to digital information." For young children, who are still developing fine motor skills and abstract thinking, the physical affordances of tangible interfaces offer several advantages over touchscreens:

- **Embodied interaction**: Physical manipulation engages motor planning, proprioception, and spatial reasoning simultaneously (Price et al., 2003).
- **Reduced cognitive load**: Physical affordances (a button looks pressable, a knob looks turnable) reduce the interface learning burden, freeing cognitive resources for content (Marshall, 2007).
- **Multimodal feedback**: Tactile confirmation (button click, encoder detent) combines with visual and auditory feedback to strengthen sensorimotor associations.

Empirical work specifically targeting preschool-age children supports these claims. Antle (2013) observed that 3–6-year-olds preferred physical–digital interactions over screen-only equivalents and sustained engagement longer; physicality, Antle argues, bridges concrete and abstract thinking in a way screens alone cannot. Manches and O'Malley (2012) showed that the deliberate, sequential nature of physical manipulation itself exercises working memory more deeply than rapid button-tapping: searching for, selecting, and placing a physical object is an extended motor–cognitive act that offloads some intermediate state into the physical arrangement, freeing mental resources for the task itself. Sylla et al. (2012) reported that tangible story tokens increased narrative complexity and sustained engagement compared with screen-only storytelling in young children.

Horn et al. (2012) make an important qualification: pure-tangible interfaces are not automatically better than graphical ones. The clearest gains come from hybrid designs that combine tangible input with rich on-screen (and on-board) feedback. Böðn's architecture follows this hybrid pattern — every interaction is physical, every response is multimodal, and the two screens are strictly used for feedback rather than as the interaction surface themselves.

2.4 Multimodal Feedback and Sensory Integration

Young children benefit from redundant, multimodal feedback—receiving the same information through multiple sensory channels simultaneously (Price et al., 2003). In Böðn, every user action produces visual (LED/screen), auditory (tone/sound), and tactile (physical click/detent) feedback. This redundancy:

- Strengthens cross-modal neural connections
- Provides confirmation even if one channel is missed
- Supports different learning preferences and sensory profiles

The specifically cross-modal case is worth drawing out, because Böðn exploits it directly. Maurer et al. (2006) replicated the classic Bouba/Kiki effect in 2.5-year-olds, demonstrating that children form non-arbitrary sound–shape associations well before literacy. Dolscheid et al. (2014) showed that even four-month-old infants map higher pitch to higher spatial position — a prelinguistic cross-modal correspondence. These capacities are the foundation for the Tone Lab's morphing blob: the shape responds to timbre (round to spiky) and the vertical position responds to pitch (low to high), giving the child a visual channel onto the same signal their ears are processing. The goal is to make the sound's structure perceptible rather than merely audible.

2.5 Auditory Processing and Musical Exposure

Early exposure to structured sound environments supports auditory processing skills that underpin language acquisition and musical development (Kraus and Chandrasekaran, 2010; Trainor, 2005). Active sound production (as opposed to passive listening) is particularly effective.

Two more specific lines of evidence inform Böðn's audio design. First, Gordon (2003)'s Music Learning Theory argues that tonal and rhythmic vocabulary should be built through open, in-tune exploration before any formal notation is introduced — children need to encounter many well-formed musical patterns before they can produce or analyse them themselves. The Tone Lab's strict-pentatonic constraint and the Sequencer's fixed tempo grid both reflect this principle: the child explores freely but everything they produce is musically well-formed. Second, Watson (1985) and Rochat (2001) argue that the perception of oneself as a *cause* of sound is strongest when input and audible effect are contingent within a narrow temporal window. Böðn's audio path is therefore engineered to keep input-to-sound latency in the low-millisecond range (the native C mixer runs on a dedicated CPU core), preserving the tight contingency loop that underwrites early agency.

2.6 Bilingualism and Executive Function

A substantial literature documents enhanced executive functions in bilingual children, attributed to the constant need to manage two active language systems (Bialystok, 2011). The bilingual advantage is most pronounced for inhibitory control and cognitive flexibility—the same skills that Böðn's game modes target.

The bilingual advantage has been contested: Paap et al. (2015) argued that positive results reflect publication bias rather than a robust effect, and Lehtonen et al. (2018)'s meta-analysis of adult studies reached a similar conclusion once publication bias was modelled. However, in the specific context of early childhood (ages 3–5), where language systems are still being established, the evidence for enhanced inhibitory control remains more consistent (Bialystok, 2011; Barac et al., 2014). The pragmatic conclusion is that dual-language exposure is worth pursuing for its intrinsic communicative value — being able to understand and speak two languages — even if the downstream executive-function benefits are smaller or more context-dependent than early work suggested. Sweden provides a particularly favourable setting for early bilingualism: English-language media exposure is pervasive, and Swedish preschools increasingly introduce English informally.

Letter Case and Early Literacy

Research on letter-name knowledge shows that preschoolers recognise uppercase letters before lowercase ones—uppercase knowledge predicts lowercase acquisition with a facilitation factor exceeding 16× (Piasta, 2014). However, since more than 95% of printed text uses lowercase forms, prolonged exposure to only uppercase delays the transition to reading real text (Treiman and Kessler, 2006). Böðn therefore uses **title case** (first letter uppercase, remainder lowercase) for all child-facing labels, giving the child the familiar uppercase anchor while building lowercase familiarity.

Dual-Language Labeling

Byers-Heinlein and Lew-Williams (2013) demonstrated that bilingual infants distinguish their two languages from birth using rhythmic and phonological cues, countering the common concern that simultaneous exposure causes confusion. Tan and Byers-Heinlein (2024) showed that *translation equivalents*—knowing a word in one language—significantly increases the probability of learning the corresponding word in the other language, especially at younger ages. A peer-reviewed synthesis of dual-language-learner research concludes that vocabulary acquisition in both languages is supported when children receive consistent, paired exposure to the same concepts in each language (Barac et al., 2014). Böðn's printed NFC cards therefore present both Swedish and English labels simultaneously (e.g. "Katt / Cat"), while the on-device UI shows one language at a time (switchable in settings) to respect the display's space constraints.

2.7 Pretend Play and Social-Cognitive Development

Lillard et al. (2013) reviewed the evidence linking pretend play to cognitive development, finding associations with theory of mind, narrative skills, and emotional understanding. While the causal direction remains debated, the correlational evidence is strong, and pretend play is universally observed across cultures during the preschool years.

Two Böðn modes are explicitly designed as scaffolded pretend-play environments. *Spaceship Cockpit* casts the child as the captain of a small ship, with the AI companion "Stellar" announcing scenarios ("Asteroids ahead! Steer out of the way!") via TTS. The physical encoders, toggles, and arcade buttons are reframed as throttle, shields, and steering — the same hardware controls become different fictional objects depending on the scenario. This kind of symbolic substitution (treating object A *as if* it were object B) is a hallmark of preschool pretend play and is thought to contribute to theory-of-mind development by requiring the child to hold two interpretations of the same object in mind simultaneously (Lillard et al., 2013). *Story Mode* complements this with branching narrative: the child guides a character through a story world by choosing between spoken options at decision points. The narrative frame supplies the pretend context; the branching structure ensures the child's choices matter. Blippa adds a third, looser register — because the interaction is goal-free, children spontaneously treat the reader as a supermarket scanner or transit gate, producing entirely child-initiated pretend scenarios that the device simply supports.

2.8 Feedback, Failure, and Growth Mindset

Dweck (2006) distinguishes between fixed mindset (ability is innate) and growth mindset (ability develops through effort). The type of feedback children receive after failure shapes which mindset they develop. Process-focused feedback ("You tried a different strategy, good!") promotes growth mindset; outcome-focused feedback ("Wrong!") promotes fixed mindset.

Böðn's feedback design follows this principle: errors produce soft, warm cues (gentle tone, dim red glow) rather than punitive signals. The intent is to frame failure as information, not judgement.

Several concrete design decisions follow from this stance. Success feedback is not louder or more dramatic than error feedback — it is simply different in tone. This is deliberate: overly celebratory success animations implicitly punish failure by contrast. No mode uses a red "wrong" buzzer, a loudly descending "sad trombone," or a shaking screen; every error is acknowledged with a soft chime and a warm, dim visual cue and then immediately gives the child another try. Simon, for example, allows up to three attempts per sequence and only grows the sequence length on success, so the child is never confronted with an impossibly long sequence. Rule Follow switches rules after several

successes rather than after errors, so the game becomes harder as the child improves rather than punishing failure with added difficulty. High-Five Friends auto-restarts after three misses rather than forcing a session-ending screen. These decisions collectively aim to keep the emotional stakes low enough that the child continues engaging — a necessary condition for the kind of repeated, voluntary practice that Diamond and Ling (2016) identify as the core ingredient of effective EF interventions.

3 System Design

3.1 Hardware Architecture

Böðn is built around an ESP32-S3 microcontroller with the following physical interface components:

- 8 mini push buttons (colour-coded, via MCP23017 I2C expander)
- 5 arcade buttons with PWM-controlled LEDs (via PCA9685)
- 2 toggle switches (via MCP23017)
- 3 rotary encoders with push buttons (navigation, volume, throttle)
- 2.8" ILI9341 TFT primary display (240×320, landscape)
- 1.8" ST7735 TFT secondary display (128×160, portrait)
- 2× WS2812 8-LED sticks + 92-LED perimeter strip (RGB)
- MAX98357A I2S amplifier with 3W speaker
- INMP441 I2S MEMS microphone
- 6600 mAh LiPo battery with thermal monitoring (DS18B20×2)
- PN532 NFC reader (I2C) for tangible card interaction

A complete bill of materials, pin assignments, and thermal-protection escalation table is maintained in the project's `docs/hardware.md`. The device enclosure is a wooden box chosen for its warmth and durability; the goal is for the object to feel like a toy rather than a piece of consumer electronics.

3.2 Software Architecture

The firmware is written in MicroPython and follows a modular architecture:

- **Rule engines** (`*_rules.py`): Pure logic modules with no hardware dependencies, testable via standard pytest on the host.
- **Screen modules** (`ui/*.py`): Render game state to displays and LEDs; handle input events.

- **Audio engine**: 16-voice native C mixer running on a dedicated CPU core, with priority channels (music, SFX pool, UI feedback) and built-in tone generators (sine, square, sawtooth, noise).
- **Session manager**: Enforces play time limits, breaks, and quiet hours.
- **i18n system**: All user-facing strings localised (Swedish default, English available).
- **Native C modules**: performance-critical paths (audio mixer, display DMA, bitmap blit, MCP23017 input scan, NeoPixel animation engine, Game of Life step kernel) are implemented as MicroPython user C modules. Each has a pure-Python fallback so the design can be iterated in the simulator before the custom firmware is built.
- **Asset resolver**: a single `resolve()` function routes all sound and TTS lookups, checking the SD card first and falling back to flash. This lets hand-recorded narration transparently override generated TTS, and keeps firmware size independent of the media library.

The emphasis on pure-logic rule engines reflects a deliberate testing strategy: every game mode's rules can be exercised under CPython on a laptop with `pytest`, while hardware-dependent code is tested separately via the Wokwi simulator or directly on the device. This keeps the developmental logic reviewable by non-embedded collaborators and dramatically shortens the iteration loop.

4 Developmental Analysis

This section maps each Böðn feature to established developmental domains and analyses coverage. Table 1 presents the full feature–domain matrix.

4.1 Feature–Domain Mapping

Each subsection below describes one of the fourteen currently shipped game modes, identifies the primary developmental aspects it targets, and links the design decisions back to the literature reviewed in Section 2.

4.1.1 Simon (Härmapa)

The Simon memory game directly targets **working memory** capacity. The starting sequence length of 2 items is calibrated to the lower bound of 4-year-old WM capacity (Diamond, 2013). Each successful round adds one item, providing adaptive difficulty within the child's zone of proximal development (Vygotsky, 1978).

4.1.2 Rule Follow (Regelleken)

Modelled after the DCCS task (Zelazo, 2006), Rule Follow alternates between "match" and "opposite" rules. The "opposite" rule requires **inhibitory control** (suppressing the prepotent matching response), while the rule switch exercises **cognitive flexibility**. The frequency of rule switches (every 4–6 correct answers) is within the range used in DCCS research.

4.1.3 Mystery Box (Mysterium)

An open-ended **cause-and-effect** exploration mode with no failure state. The design follows the "guided play" model (Weisberg et al., 2013): the environment is structured (specific button–colour mappings, eight hand-picked "magic" combinations) but the child's exploration is entirely self-directed. A persistent 16-tile recipe book on the secondary display marks each single cap and each magic pair as it is found, making the unexplored space visible without prescribing an order—a soft carrot consistent with Bonawitz et al. (2011) on exploration-driven causal learning. The three modifiers (colour invert, LED mirror, hue rotation) are themselves gated as unlocks at 5, 10, and all-eight-singles discoveries, so the toggles and encoder feel earned rather than mystery hardware. Incidental working memory ("which two haven't I tried?"), cognitive flexibility (modifier unlocks reframe the rules mid-play), and persistence (collect-them-all carrot) are present but secondary to the open-exploration core.

4.1.4 Flöde

A **spatial reasoning** and **problem-solving** puzzle that requires aligning gaps in vertical walls. The snap-to-grid mechanic reduces motor precision demands while preserving the cognitive challenge.

4.1.5 Garden of Life (Trädgården)

A **pattern recognition** mode based on Conway's Game of Life. The child plants coloured seeds and observes emergent behaviour—an implicit introduction to complex systems and self-organisation. The mode exercises no executive functions but provides rich visual stimulation and cause-and-effect learning.

4.1.6 Soundboard (Ljudbräda)

An **auditory processing** and **open exploration** mode. Customisable sound banks allow parents to create themed experiences. The volume encoder provides fine motor practice with immediate auditory feedback.

4.1.7 Spaceship Cockpit (Rymdskepp)

The most developmentally comprehensive mode, targeting **cognitive flexibility** (5 scenario types requiring different inputs), **listening comprehension** (TTS-driven instructions), and **imaginative play** (structured pretend-play as a spaceship captain). The adaptive difficulty system adjusts timeout windows based on the child's recent performance, maintaining an appropriate challenge level.

4.1.8 Story Mode (Saga)

A data-driven branching narrative engine where the child listens to narrated scenes and makes choices via arcade buttons to steer the plot. Primary targets: **listening comprehension** (must listen to narration to choose), **bilingual exposure** (full narration in both Swedish and English), and **imaginative play** (guiding a character through a narrative world). Stories are defined as script files on the SD card; adding new stories requires no code changes. The first story adapts Beatrix Potter's *The Tale of Peter Rabbit* (public domain) into a choose-your-own-adventure with four endings.

4.1.9 High-Five Friends (Heja-kompisar)

A reaction-time game targeting **hand-eye coordination** and **inhibitory control**. Animals "pop up" on illuminated arcade buttons and the child must tap the lit button before it disappears. The reaction window shrinks from 2 s to 0.5 s as the child succeeds, providing adaptive difficulty within the child's zone of proximal development (Vygotsky, 1978). Gentle miss feedback (soft sound, dim animation) and automatic restart after three misses support **persistence** while keeping emotional stakes low (Dweck, 2006). The mode also exercises **auditory processing** (distinct hit/miss sounds) and **self-regulation** (managing the frustration of near-misses).

4.1.10 Sortera — NFC Classification Game

A tangible implementation of the Dimensional Change Card Sort (DCCS) task (Zelazo, 2006), targeting **cognitive flexibility** and **inhibitory control**. The device announces a sorting rule ("Find the animals!"), and the child scans NFC cards. After several correct matches the rule switches to a different dimension ("Find all the red ones!"), requiring the child to inhibit the previous sorting strategy. Sortera is the first mode to provide **active bilingual reinforcement**: on each correct scan, the screen displays both the Swedish and English word (e.g. "Katt / Cat") matching the physical card, and bilingual TTS announces both words—primary language first, then secondary (Tan and Byers-Heinlein, 2024). This dual-label approach leverages the translation-equivalent bootstrapping effect while keeping game instructions in the configured language for clarity.

4.1.11 Räkna — Progressive Math with Tangible Numbers

A six-level NFC math mode targeting **pattern recognition** (subitising), **sequential thinking** (counting and equation order), **problem solving** (composing equations step by step), and **fine motor** coordination (handling multiple cards per equation). Räkna implements Bruner's Concrete–Pictorial–Abstract progression (Bruner, 1966; Willingham, 2017): a physical dot-pattern card is placed on the reader (concrete), the screen renders matching dots (pictorial), and from Level 4 onwards the numeral and operator cards are introduced (abstract). Dice-face dot patterns at low levels rather than numerals reflect the developmental evidence that subitising is the foundation of number sense, not numeral recognition (Clements and Sarama, 2014). Levels span free exploration (ages 3–4), quantity matching and more/less comparison (4–5), addition and subtraction with tokens (5–6), and symbolic equations (6–7+). A left-to-right linear number path on the display implements Siegler and Ramani's (2009) finding that board-game-style number lines measurably improve preschool number competence. Bilingual TTS announces each quantity in Swedish and English, carrying the translation-equivalent bootstrapping effect (Tan and Byers-Heinlein, 2024) into a cognitively engaging math task. The mode is always self-paced and never timed — a deliberate design choice given the evidence that timed math exercises produce anxiety even in preschoolers. Together with Sortera, Räkna addresses the early-math domain that Duncan et al. (2007) identified as the single strongest predictor of later academic achievement.

4.1.12 Blippa — Free-Play NFC Discovery

A goal-less, failure-less companion to the other NFC modes targeting **open exploration**, **cause and effect**, and **imaginative play**. Tapping any owned card—from any card set the device knows about— produces a full-screen emoji, a bilingual label, and either a procedural POS "blip" or an optional per-card sample. There is no rule to follow, no score, and no wrong scan: the mode exists specifically to turn a growing collection of physical cards into a low-friction exploration toy. This design is motivated by evidence that children's causal learning is strongest when exploration is self-directed rather than instructed (Bonawitz et al., 2011), and it intentionally creates a surface for the cashier and transit-gate pretend play that 3–5-year-olds gravitate to, linking open exploration with imaginative play. Unknown card identifiers within a known card set produce a deterministic "mystery blip" whose pitch is derived from a stable hash of the id, so a given card always sounds the same—reinforcing object permanence and rehearsal of cause-effect contingencies.

4.1.13 Sequencer (Sequenser)

A loop-based step sequencer targeting **working memory**, **sequential thinking**, **pattern recognition**, and **creative expression**. Five percussion tracks (arcade buttons) and one melody track (eight mini buttons mapping to a pentatonic scale) are recorded into an 8- or 16-step grid during looped playback. Toggle switches control play/pause, step count, and metronome—making abstract musical concepts tangible. The mode introduces computational thinking through creative play (Papert, 1980; Bers, 2018): building patterns, running them, hearing something unexpected, and debugging by toggling individual steps. BPM is adjustable via encoder (70–140), providing fine motor practice with immediate auditory feedback.

4.1.14 Tone Lab (Ljudlabb)

A real-time synthesiser sandbox targeting **auditory processing**, **cause-and-effect learning**, **visual processing via cross-modal binding**, and **creative expression**. Two encoders control pitch (strict pentatonic over two octaves) and timbre (a five-step morph from smooth sine to fuzzy noise). Five arcade buttons snap to fixed pentatonic pitches so nothing ever sounds "wrong"—consistent with Gordon's Music Learning Theory (Gordon, 2003), which argues tonal vocabulary is built through open but in-tune exploration. Eight mini buttons provide hold-to-apply effects (vibrato, tremolo, pitch bend up/down, octave jump up/down, stutter, harmony); each effect is a single reversible gesture, keeping the cause-and-effect link transparent (Watson, 1985). All effects run inside the native audio mixer, so every input change is reflected in the waveform within milliseconds—preserving the tight contingency loop that young children need to perceive themselves as the cause of the sound (Rochat, 2001).

Two visual channels onto the same signal reinforce cross-modal learning. A live oscilloscope shows the actual waveform, and a central blob morphs from round (sine) to spiky (noise) in lockstep with the timbre encoder— exploiting the Bouba/Kiki correspondence first documented by Köhler (1929) and replicated in toddlers by Maurer et al. (2006). The blob's vertical position tracks pitch (higher tones higher on screen), drawing on the pitch-height correspondence present even in infants (Dolscheid et al., 2014). A toggle switch swaps the big oscilloscope between the primary and secondary displays so older children or accompanying adults can inspect the waveform directly while the child remains focused on the expressive blob.

Table 1: Feature–developmental domain coverage matrix. ● ● ● = primary target; ●● = significant; ● = incidental; blank = not applicable.

Domain	*Simon*	*Rule Follow*	*Mystery Box*	*Flöde*	*Garden*	*Soundboard*	*Spaceship*	*Story Mode*	*High-Five*	*Sequencer*	*Tone Lab*	*Se*
Working memory	● ● ●	●●	●	●			●●	●●	●	● ● ●	●	
Inhibitory control	●	● ● ●					●●		●●			●
Cognitive flexibility		● ● ●	●				● ● ●			●●	●	●
Cause & effect			● ● ●		●●	●●		●●	●●	●●	● ● ●	●
Pattern recognition	●●		●●		● ● ●					● ● ●	●	●
Spatial reasoning				● ● ●								
Sequential thinking	● ● ●		●	●●				●●		● ● ●		
Problem solving				● ● ●			●			●●		
Fine motor	●	●	●	●●	●	●	●●	●	●	●●	●●	●
Hand–eye coord.	●●	●●		●●			●●		● ● ●	●●	●	
Auditory processing	●●					● ● ●	●●	● ● ●	●●	●●	● ● ●	●
Visual processing	●●	●●	● ● ●	●●	● ● ●		●●	●●	●●	●●	●●	●
Listening comp.						●	● ● ●	● ● ●				
Bilingual exposure	●	●		●	●	●	●	● ● ●	●	●	●	●
Self-regulation	●●	● ● ●			●		●●	●	●●			●
Persistence	●●	●	●	● ● ●			●	●	●●	●●		
Imaginative play							● ● ●	● ● ●		●●	●	
Open exploration			● ● ●		● ● ●	● ● ●					● ● ●	
Divergent thinking			● ● ●		●●					● ● ●	●●	
Creative expression					●●	● ● ●				● ● ●	● ● ●	

4.2 Coverage Matrix

4.3 Gap Analysis

The current fourteen-mode catalogue yields strong coverage (three or more features at primary or significant level) across every developmental aspect except spatial reasoning. The shift is the result of several deliberate additions that each filled a specific gap identified in earlier revisions of this matrix. Story Mode gave listening comprehension, bilingual exposure, and imaginative play a second primary-level feature apiece. The Sequencer introduced creative expression and computational thinking. High-Five Friends added dedicated hand–eye coordination with adaptive timing. The three NFC modes then added a tangible manipulation layer: Sortera provides a tangible Dimensional Change Card Sort (Zelazo, 2006) strengthening cognitive flexibility and inhibitory control; Räkna opens the early-mathematics domain that Duncan et al. (2007) identified as the single strongest predictor of later academic achievement, following Bruner's Concrete–Pictorial–Abstract progression (Bruner, 1966) and Siegler and Ramani's (2009) linear-number-board evidence; and Blippa closes the tangible cause-and-effect loop without any goal or failure state, aligning with Bonawitz et al. (2011) on exploration-driven causal learning. Finally, Tone Lab added a third primary-level auditory-processing mode alongside Soundboard and Story Mode and—uniquely—targeted cross-modal sound–shape perception (Köhler, 1929; Maurer et al., 2006; Dolscheid et al., 2014), a developmental capacity no previous mode addressed directly.

The single remaining under-covered domain is:

1. **Spatial reasoning**: still primarily Flöde, with Räkna's left-to-right number path contributing only incidentally. A tangram or construction mode would add breadth, and the predictive link between preschool spatial assembly and later mathematics (Verdine et al., 2014) makes this a high-leverage gap to close.

Two further areas remain thinner than the rest even though they pass the three-feature threshold: problem solving is presently Flöde (primary), Sequencer (debugging), and Räkna (equation building), so a scavenger-hunt NFC puzzle mode would usefully broaden it; and listening comprehension rests on only two primary features (Spaceship, Story Mode), so a dedicated language-switching game could both strengthen it and further leverage bilingual exposure for EF development (Bialystok, 2011).

5 Discussion

5.1 Tangible Interfaces vs. Touchscreen Applications

The dominant form factor for children's digital learning tools is the tablet-based app. Böðn's commitment to physical controls is a deliberate departure, grounded in three specific arguments.

First, the embodied-cognition literature holds that thinking in young children is scaffolded by the body: the actions a child can perform constrain and shape the concepts they can form (Antle, 2013; Price et al., 2003). A button that clicks, an encoder that detents, and a toggle that latches give the child proprioceptive feedback independent of the screen — the hand *knows* something happened even before the eyes register it. Touchscreen taps provide none of this. Second, physical controls act as an external memory (Manches and O'Malley, 2012): an NFC card on the table is visible in the peripheral field whether or not the child is currently attending to it, which supports the kind of extended planning behaviour that a rapidly scrolling touchscreen actively disrupts. Third, and pragmatically, physical devices are easier to regulate: a wooden box with a power switch is harder to use compulsively than an app embedded in a general-purpose device that also contains YouTube.

Commercial tablet-based EF training programs (Cogmed and similar) have produced mixed results in preschool populations, with transfer effects narrow to the trained tasks (Diamond and Ling, 2016). Böðn does not attempt to produce drill-based training; it instead embeds EF demands inside games the child chooses to play repeatedly, in line with the broader-transfer approaches Diamond & Ling identified as effective.

5.2 Adaptive Difficulty and the Zone of Proximal Development

Vygotsky (1978) proposed that learning is most efficient when a child is supported to operate just beyond what they can already do unaided — the Zone of Proximal Development (ZPD). Several Böðn modes implement explicit adaptive-difficulty systems that aim to keep the child in this zone. Simon grows sequence length by one item per success and never shortens it unless the child fails repeatedly, so the demand tracks the child's capacity. Spaceship shortens the response timeout as the child succeeds; a child who is struggling receives more time, a child who is mastering the scenarios receives less. High-Five Friends shrinks the reaction window from 2 s down to 0.5 s across a round. Räkna's six-level progression makes the same principle explicit at a coarser granularity: the child begins with free dot-card exploration and moves to symbolic equations over months or years as readiness emerges.

An implicit assumption of the ZPD framing is that the *right* level of challenge is knowable in advance. In practice, Böðn's adaptive loops use simple rules based on re-

cent success/failure counts rather than any formal model of the child's skill. This is adequate for single-session adaptation but does not persist across sessions. A Bayesian skill-estimation layer is listed in Section 6 as a plausible extension.

5.3 Feedback Design and Growth Mindset

The growth-mindset literature (Dweck, 2006) argues that feedback focused on process ("you tried a different strategy — good") produces more persistent learners than feedback focused on outcome or innate ability. Böðn operationalises this at the level of every error sound and every animation: mistakes produce soft, warm cues; successes produce differently-shaped cues rather than proportionally larger ones; and no mode ever displays a "game over" screen in large red letters. This is an unusually strong stance — many commercial children's games use loud failure feedback explicitly because it provokes the "one more try" response — and the trade-off is accepted deliberately: Böðn prioritises emotional safety over short-term engagement loops, betting that lower-frustration play sustains longer-term engagement (Diamond and Ling, 2016).

A subtler design decision concerns achievements and progression markers. Böðn has no stars, no points, no leaderboards, and no unlockable content gated behind performance. The rationale is two-fold: first, extrinsic rewards can undermine intrinsic motivation in young children; second, rewards create winners and losers in a single-device household, which is undesirable. Level progression in Räkna, for example, is surfaced only in the parental controls, not to the child.

5.4 Session Limits and Healthy Technology Use

The WHO (World Health Organization, 2019) recommends no more than one hour of sedentary screen time per day for children aged 2–4. The key word is *sedentary*: the concern is not the photons emitted by a screen but the passive consumption pattern that typically accompanies them. Böðn challenges the category: its screens exist, but they respond only to physical action, and the child cannot progress without moving. This is closer to "physical play with visual feedback" than to traditional screen time, and the session-limit framework is sized accordingly.

Böðn's parental controls allow configurable session duration, daily session count, mandatory breaks between sessions, and quiet hours. Session-end is signalled by a gradual wind-down animation rather than an abrupt cutoff — the child learns that "time to stop" is a predictable, normal part of play, rather than a punishment. All play-history data stays on the device; no telemetry is uploaded to a cloud, and the parental web dashboard is served directly from the ESP32 over the local network.

5.5 Limitations

Several limitations should be acknowledged. First, no formal user studies or controlled evaluations have been conducted; the claims in this paper are design-rationale arguments grounded in the developmental literature, not experimental findings about Böðn itself. Second, Böðn is currently a single-user device: every mode assumes one child at a time, and cooperative or turn-taking features are not yet implemented. This limits the system's ability to support socio-cognitive development in peer contexts. Third, the bilingual-advantage literature that motivates some of Böðn's design decisions is actively contested (Paap et al., 2015; Lehtonen et al., 2018); the dual-language features should be viewed primarily as supporting communicative competence in both languages, with downstream EF effects as a possible but uncertain bonus. Fourth, hardware constraints (low-resolution displays, no haptic feedback beyond the physical controls, a modest 3 W speaker) limit the range of sensory experiences the device can deliver. Finally, the author is not a developmental psychologist; the literature review draws on published work but has not been peer-reviewed by domain experts.

6 Future Work

1. **NFC tangible interaction layer**: Integrate a PN532 NFC reader to enable tagged physical objects (cards, stickers, figurines) as input. Research on tangible user interfaces consistently shows that physical manipulation benefits learning in preschoolers through reduced cognitive load, embodied engagement, and support for classification tasks (Marshall, 2007; Antle, 2013; Manches and O'Malley, 2012). Planned NFC modes include:

    - *Sortera* — a tangible classification game implementing the DCCS task (Zelazo, 2006) with physical cards, targeting cognitive flexibility and inhibitory control.
    - *Saga Builder* — tangible storytelling where scanned character/setting cards trigger narration, extending the existing Story Mode with a richer physical input method (Sylla et al., 2012).
    - *Vocabulary Explorer* — bilingual word cards for active language switching, leveraging the bilingual EF advantage (Bialystok, 2011).
    - *Memory Match* — a physical search-and-select task that adds motor planning to working memory challenges (Manches and O'Malley, 2012).
    - *Räkna* — a progressive math mode using dot-pattern NFC cards as tangible number tokens. Follows Bruner's Concrete-Pictorial-Abstract progression (Bruner, 1966): physical tokens (concrete) → dot displays (pictorial) → numerals (abstract). Six levels spanning subitising, counting, comparison, addition, and subtraction (ages 3–7). Grounded in evidence that early math is

the strongest predictor of later academic achievement (Duncan et al., 2007), that linear number board games produce measurable and persistent gains in preschoolers' number competence (Siegler and Ramani, 2009), and that kindergarten number sense predicts math achievement through third grade (Jordan et al., 2009).

The NFC interaction pattern (find object → bring to reader → hold) gives children physical agency in a way that fixed buttons cannot, and has been commercially validated by products like Toniebox and Yoto Player for children as young as 3. See the companion document `nfc_tangible_learning.md` for the full research summary.

2. **Empirical validation**: Conduct pre–post studies comparing EF measures (DCCS, digit span) in children using Böðn vs. control activities.

3. **Record & Replay mode**: Implement voice recording and playback with effects (pitch shift, echo) to strengthen creative expression, auditory processing, and bilingual exposure.

4. **Cooperative play**: Add turn-taking and collaborative modes to support social-cognitive development.

5. **Adaptive difficulty modelling**: Implement Bayesian difficulty estimation to maintain each child in their zone of proximal development across play sessions.

6. **Longitudinal tracking**: Use the existing session logging infrastructure to track developmental progress over months and years.

7. **Active bilingual features**: Develop language-switching games that leverage the bilingual EF advantage.

7 Conclusion

We have argued that tangible, open-source hardware can deliver developmentally informed play experiences for young children that neither screen-based apps nor traditional toys match alone. Böðn's design grounds each feature in established developmental science: executive functions are exercised by Simon, Rule Follow, Sortera, and Sequencer; early mathematics by Räkna; bilingual exposure by Story Mode, Sortera, Räkna, and Blippa; creative expression by Tone Lab and Sequencer; pretend play by Spaceship and Story Mode. Across all fourteen modes a small set of recurring design principles do the heavy lifting — physical-first interaction with multimodal feedback, guided play with preserved child agency, low-stakes failure, and contingent input–output loops tight enough to support the perception of oneself as a cause.

The coverage analysis (Section 4.2) indicates that the current catalogue addresses nearly every developmental aspect identified as relevant for ages 3–7, with spatial reasoning as the one remaining gap. Future work should close that gap, add empirical validation studies, and extend the platform to cooperative play. Because Böðn is fully open source — hardware designs, firmware, documentation, and the research behind the design decisions — we hope the approach can serve both as a platform for empirical work in child development and as an invitation to other makers, educators, and developmental researchers to collaborate on physical computing for young children.

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